

Prelims

For this HW2 I (Frederik Hüttemann, 108021215247) cooperated with my colleague (Ben Olschar, 108021211678). We did implement the code on our own but we mutually checked each others code. He also uploaded his code + results as we didn't know if uploading one result is sufficient for both.

Results

To generate the results, the code was run on my GPU (RTX 4060 Dual). To avoid recoding noise, the tests were run for 10.000 times.

Results for different matrix sizes:

Matrix Size: 10x10, Block Size: 16x16 => CPU Avg Time: 0.237513 ms, GPU Avg Time: 0.048213 ms

Matrix Size: 100x100, Block Size: 16x16 => CPU Avg Time: 23.398304 ms, GPU Avg Time: 0.048256 ms

Matrix Size: 500x2000, Block Size: 16x16 => CPU Avg Time: 2505.173993 ms, GPU Avg Time: 0.115916 ms

Matrix Size: 1000x1000, Block Size: 16x16 => CPU Avg Time: 2555.434394 ms, GPU Avg Time: 0.112378 ms

Matrix Size: 100x10000, Block Size: 16x16 => CPU Avg Time: 2617.185593 ms, GPU Avg Time: 0.108724 ms

Results for different block sizes:

Matrix Size: 100x10000, Block Size: 16x16 => GPU Avg Time: 0.083040 ms

Matrix Size: 100x10000, Block Size: 16x32 => GPU Avg Time: 0.083567 ms

Matrix Size: 100x10000, Block Size: 32x16 => GPU Avg Time: 0.080189 ms